

MIDTERM I—SUGGESTED SOLUTIONS

ECON498—ADVANCED TOPICS IN ECONOMICS: GENERAL EQUILIBRIUM

APRIL 5, 2026

Note: these are my *suggested* solutions, with the keyword being *suggested*. There are several ways to prove the same result. I will gladly welcome requests for clarifications or to discuss other possible answers.

1. (10 points each) Prove or disprove the following two statements.

- (a) For any two sets S and T , $\text{co}(S \cup T) = \text{co}(S) \cup \text{co}(T)$.
- (b) If S is convex, then $\bigcup_{x \in S} B_\varepsilon(x)$ is convex for every $\varepsilon > 0$.

Solution

- (a) False. Let $S = \{1\}$ and $T = \{2\}$.
- (b) True. This follows from the observation that $\bigcup_{x \in S} B_\varepsilon(x) = S + B_\varepsilon(0)$, which appears as an exercise in the notes, and the result that the vector sum of convex sets is convex, which also appears as an exercise in the notes. ($\triangleright \wedge \top \wedge \triangleleft$)

2. A set $S \subseteq \mathbb{R}^n$ is a cone if $\lambda s \in S$ whenever $s \in S$ and $\lambda \in [0, \infty)$. For every set S , define the set

$$S^* = \{ p : p \cdot s \leq 0 \text{ for every } s \in S \}.$$

It is easily observed that for every S , S^* is a closed and convex cone. Accept the following two statements as correct.

-  If S is a cone and if there exist a vector p and a scalar c such that $p \cdot s < c$ for every $s \in S$, then $c > 0$ and $p \cdot s \leq 0$ for every $s \in S$.
-  If S is a closed and convex cone, then $(S^*)^* = S$.

Consider the following statement.

Let S be a closed and convex cone and K be a compact and convex set. Then

$$K \cap S^* \neq \emptyset \iff \forall s \in S \exists k \in K : s \cdot k \leq 0.$$

- (a) (15 points) Prove this result using statements  and .
- (b) (5 points) Prove statement .
- (c) (10 points) Prove statement  using statement .

Solution

- (a) ($\Rightarrow$) Let $z \in K \cap S^*$. Then, by definition, $z \cdot s \leq 0$ for all $s \in S$. ($\Leftarrow$) We proceed by contrapositive. If $K \cap S^* = \emptyset$, then by the Strong Separation Theorem, there exists $q \neq 0$ such that for every $(k, p) \in S^* \times K$, such that $p \cdot q < q \cdot k$. Choose c such that $p \cdot q < c$ for all $p \in S^*$ and $c < q \cdot k$ for all $k \in K$. Since S^* is a cone, this implies, by , that $c > 0$ and $p \cdot q \leq 0$ for every $p \in S^*$. Thus, $q \in (S^*)^*$, which, by , is S . Thus, we have found $q \in S$ such that $q \cdot k > 0$. $\wedge \circ \circ \wedge$
- (b) *Proof of* . Since S is a cone, $0 \in S$ (take any $s \in S$ and set $\lambda = 0$). Evaluating the bound at the origin gives $p \cdot 0 = 0 < c$, so $c > 0$. We now prove $p \cdot s \leq 0$ for every $s \in S$ by contrapositive. Suppose $p \cdot s > 0$ for some $s \in S$. Since S is a cone, $\lambda s \in S$ for every $\lambda \geq 0$, and $p \cdot (\lambda s) = \lambda(p \cdot s)$. For every $d > 0$ we may choose $\lambda = d/(p \cdot s)$ to obtain $p \cdot (\lambda s) = d$. In particular, since $c > 0$, there exists $s' \in S$ with $p \cdot s' \geq c$, contradicting the bound. We therefore must have $p \cdot s \leq 0$ for every $s \in S$. $\wedge \circ \circ \wedge$
- (c) *Proof of* . If $S = \{0\}$, then there is nothing to prove, for $S^* = \mathbb{R}^n$ and $(S^*)^* = \{0\}$. Suppose that $S \neq \{0\}$. If $s \in S$ and $p \in S^*$, then $p \cdot s \leq 0$, implying that $s \in (S^*)^*$. Thus, $S \subseteq (S^*)^*$. Now suppose that $x \in (S^*)^*$. Towards a contradiction, suppose that $x \notin S$. Since S is a non-empty, closed and convex set, then we can apply the Strong Separation Theorem to find $q \neq 0$ such that $q \cdot s < q \cdot x$ for all $s \in S$. But by , $q \cdot x > 0$, and $q \cdot s \leq 0$ for every $s \in S$; Thus $q \in S^*$. The previous statements together with $x \in (S^*)^*$ imply that $q \cdot x \leq 0$, which is a contradiction. Thus we have that $(S^*)^* \subseteq S$. $\wedge \circ \circ \wedge$

3. (20 points) Let $X \subseteq \mathbb{R}_+^m$ be nonempty and compact, and fix a vector $e \in X$. Interpret X as the consumption set and e as the endowment of a consumer. Define the consumer's budget correspondence¹ $\beta : \mathbb{R}_{++}^m \rightrightarrows X$ by

$$\beta(p) = \{x \in X : p \cdot x \leq p \cdot e\}$$

for every $p \in \mathbb{R}_{++}^m$. Prove or disprove the following statement.

The budget correspondence is lower hemicontinuous.

Solution

This statement is not correct. Consider the following example.

$$m = 2$$

$$X = [0, 1] \times [0, 1] \setminus \{(x_1, x_2) : \frac{1}{3} < x_2 < \frac{2}{3}\}$$

$$e = (1, 0)$$

We will show that β is not lhc at $p = (2, 3)$. Observe that $\beta(2, 3) = A \cup \{(0, \frac{2}{3})\}$ where $A = \text{co}\{(0, 0), (1, 0), (0, \frac{1}{3}), (\frac{1}{2}, \frac{1}{3})\}$. For any $0 < \delta < \frac{1}{3}$, the set

$$U = B_\delta\left((0, \frac{2}{3})\right) \cap X$$

is open in X and has nonempty intersection with A . For every $\varepsilon > 0$, pick $(2, 3 + \frac{\varepsilon}{2}) \in B_\varepsilon(p) \cap \mathbb{R}_{++}^m$ and observe that the budget set

$$\beta(2, 3 + \frac{\varepsilon}{2}) = \{(x_1, x_2) \in X : 2x_1 + (3 + \frac{\varepsilon}{2})x_2 \leq 2\}$$

is completely contained in A and therefore has empty intersection with U . Thus β is not lhc at $p = (2, 3)$.

4. (20 points) The following result is known as the **Intermediate Value Theorem** or **Bolzano's Theorem**, which hopefully you learned in Calculus I.

If $f : [a, b] \rightarrow \mathbb{R}$ is continuous, and if y lies between $f(a)$ and $f(b)$, i.e., if $f(a) \leq y \leq f(b)$ or $f(b) \leq y \leq f(a)$, then there exists some $x \in [a, b]$ such that $f(x) = y$.

Prove that the Intermediate Value Theorem implies **Brouwer's Fixed Point Theorem** when $m = 1$. (If $K \subseteq \mathbb{R}^m$ is nonempty, compact and convex and if $f : K \rightarrow K$ is continuous, then f has a fixed point.)

Solution

Firstly, notice that a non-empty, compact, and convex set in $\mathbb{R}$ is of the form $[a, b]$. Thus, the result that we aim to prove is: *If $f : [a, b] \rightarrow [a, b]$ is continuous, then f has a fixed point.* Let $g : [a, b] \rightarrow \mathbb{R}$ be given by $g(x) = f(x) - x$ for all $x \in [a, b]$. The function g is continuous, and since $f([a, b]) \subseteq [a, b]$, we have that $a \leq f(x) \leq b$ for all $x \in [a, b]$, and thus, $g(a) = f(a) - a \geq 0$ and $g(b) = f(b) - b \leq 0$. Then g satisfies the hypotheses of Bolzano's theorem, so there must exist $x^* \in [a, b]$ with $g(x^*) = 0$ implying that $f(x^*) = x^*$. ⁹

5. Let $K \subseteq \mathbb{R}^\ell$ be a nonempty closed set. The *support function* of K is the function $\mu_K : \mathbb{R}^\ell \rightarrow \overline{\mathbb{R}}$ defined² by

$$\mu_K(p) = \inf_{x \in K} p \cdot x.$$

For each $p \in \mathbb{R}^\ell \setminus \{0\}$ and $\alpha \in \mathbb{R}$, write

$$H^+(p, \alpha) = \{x \in \mathbb{R}^\ell : p \cdot x \geq \alpha\}$$

for the closed half-space induced by normal vector p and scalar α . We also set $H^+(0, \alpha) = \mathbb{R}^\ell$ if $\alpha \leq 0$ and $H^+(0, \alpha) = \emptyset$ if $\alpha > 0$.

Fact (accept without proof). Let $f : \mathbb{R}^\ell \rightarrow \overline{\mathbb{R}}$ be concave. A vector $v \in \mathbb{R}^\ell$ is a *supergradient* of f at $\bar{p}$ if

$$f(p) \leq f(\bar{p}) + v \cdot (p - \bar{p}) \quad \text{for all } p \in \mathbb{R}^\ell.$$

The *subdifferential* of f at $\bar{p}$ is the set of all supergradients, denoted $\partial f(\bar{p})$. Then f is differentiable at $\bar{p}$ if and only if $\partial f(\bar{p})$ is a singleton $\{v\}$, in which case $\nabla f(\bar{p}) = v$.

- (a) (5 points) Show that μ_K is **concave** and **homogeneous of degree one**, i.e. $\mu_K(\lambda p) = \lambda \mu_K(p)$ for all $\lambda \geq 0$.

¹Recall that $\mathbb{R}_+ = [0, \infty)$ and $\mathbb{R}_{++} = (0, \infty)$.

²Recall that $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$.

(b) (10 points) Suppose K is convex. Show that

$$K = \bigcap_{p \in \mathbb{R}^\ell} H^+(p, \mu_K(p)).$$

Hint: one inclusion is immediate from the definition of μ_K . For the other, use a Separation Theorem.

(c) (5 points) Show that

$$\partial \mu_K(\bar{p}) = \{\bar{x} \in K : \bar{p} \cdot \bar{x} = \mu_K(\bar{p})\}.$$

That is, the subdifferential of μ_K at $\bar{p}$ is exactly the set of minimizers of $p \cdot x$ over K , evaluated at $\bar{p}$.

(d) (5 points) Using parts (a)–(c) and the Fact above, prove the following.

Theorem (The Duality Theorem). Let $K \subseteq \mathbb{R}^\ell$ be a nonempty closed set and let μ_K be its support function. Then there exists a unique $\bar{x} \in K$ such that $\bar{p} \cdot \bar{x} = \mu_K(\bar{p})$ if and only if μ_K is differentiable at $\bar{p}$. Moreover, in this case,

$$\nabla \mu_K(\bar{p}) = \bar{x}.$$

(e) (15 points) Let $K \subseteq \mathbb{R}^\ell$ now be an arbitrary nonempty closed set (not necessarily convex). Show that

$$\bigcap_{p \in \mathbb{R}^\ell} H^+(p, \mu_K(p)) = \overline{\text{co}(K)}.$$

Conclude that $\mu_K = \mu_{\overline{\text{co}(K)}}$, i.e. the support function cannot distinguish K from its closed convex hull.

(f) (20 points) Let $f : \mathbb{R}_+^\ell \rightarrow \mathbb{R}_+$ be a production function that is **strictly quasiconcave** and **continuous**. Given an output level $q > 0$ and input prices $w \in \mathbb{R}_{++}^\ell$, the *cost function* and *conditional input demand* are defined by

$$c(w, q) = \min_{x \geq 0} \{w \cdot x : f(x) \geq q\}, \quad x(w, q) = \arg \min_{x \geq 0} \{w \cdot x : f(x) \geq q\}.$$

Use the Duality Theorem to prove **Shephard's Lemma**:

$$\nabla_w c(w, q) = x(w, q).$$

Hint: identify $c(w, q)$ as the support function of an appropriate set, and argue that $x(w, q)$ is unique.

6. (10 points) Below is a picture of Panqueque, General Equilibrium aficionado, Dragonborn,³ and Cat-Emperor of the Universe.



Panqueque being mysterious, circa Winter 2025.

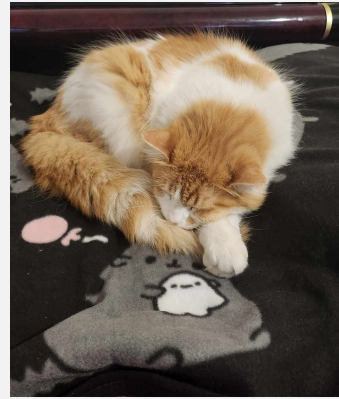
Panqueque has provided emotional support in constructing the lecture notes, from his base of operations in Mexico. Based on your knowledge of Panqueque and on his mysterious stance above, what is Panqueque's **favorite result or theorem** covered in this course? **Justify your answer.**

Note: answers such as "he has no favorite theorem, he is a cat" will receive zero points. Panqueque is a connoisseur and will be grading this question himself.

³A being with the soul of a dragon.

Solution

Panqueque has been observed to consistently seek out convex sets, most notably blankets, and position himself in their interior, thereby achieving a fleeting but meaningful separation from the horrors of the outside world. This behavior is consistent with a deep appreciation for the Strict Separation Theorem. On the other hand, Panqueque is known to be coverable by a ball of fixed radius in the appropriate metric, which, combined with his predilection for closed and bounded configurations, suggests an equally plausible affinity for the Heine–Borel Theorem. The professor acknowledges that interpreting the will of Panqueque is an inherently uncertain exercise, and that full credit may be awarded for well-structured arguments concerning Panqueque’s mathematical predilections. ($=^{\wedge}\bullet\omega\bullet^{\wedge}=$)



Panqueque, inside a closed and convex set, isolating himself from the horrors of this cruel world, circa Summer 2025.